Support copilot demo

Markus Isaksson · Associate AI Engineer interview · Wednesday 11 November 2026, 15:00-15:45 (Stockholm)

A short note on the Swedish support-copilot demo. All data is synthetic. Nothing is sent unless a person approves, edits, rejects or takes over a case.

Public repository: https://github.com/MarkusIsaksson1982/cambio-support-copilot

What it is

A fail-closed pipeline for healthcare-software support staff:

raw ticket → mask personal data → rule-first route → at most one narrow model step → schema validate → human approval or handoff → JSONL audit (masked)

The live page is `GET /` in `cambio-support-copilot`, running on adapter `none` (the same offline path as the evaluation pack). Inspelat mode reads four frozen examples and does not post a decision. `awaiting_approval` means a draft is waiting. It does not mean approval has already started. `sent` is the local audited label for approve or edit. It is not delivery to an external ticketing system. There is no ticketing integration in this demo.

How this was built

I built this with AI-assisted coding and agentic workflows, not by hand-writing most of the code. I directed the work: I set the scope, designed the safety gates and evaluations, reviewed diffs, decided what to keep or reject, and verified the test suite and the four frozen examples.

The tools were Grok Bot (orchestration), Grok Build, Codex, OpenCode, Cursor and local models. Prompts sent to outside models were checked first so they contained no personal data or tokens.

What is true now

- Copilot suite: 137 passed, 10 xfailed. The xfails are deliberate: Luhn-invalid personnummer are not rewritten and still force a handoff. Slash signatures such as `/Mikael` and the placeholder `patient X` are masked. A bare given name with no slash, title or surname can still appear.
- Offline evaluation, adapter `none`, 44 synthetic tickets, 3 identical repeats. A live model is not in these numbers: schema 44/44, audit 44/44, handoff 39/44, awaiting approval 5, patient-data 8/8, needs_human 8/8, retrieval hits 13/44, faithfulness 12/12 cited drafts, recall 12/12 retrieved non-journalfel tickets, incident chains 4/4, repeat mismatches 0.
- Retrieval drafts and the incident plugin are in the product. An incident always hands off. The short demo does not open the incident form.
- Golden labels were set for 16 rule-voter disagreements. One safety flag was raised: a ticket about a wrong patient id in HL7 now needs a human, while the category stays integration.
- A small live-model probe (three repeats of two tickets on one free OpenCode model) still handed off the patient-data case and the no-KB case, while labels moved. That probe is not the evaluation score. The interview path stays on adapter `none`. Details: `docs/LIVE-PROBE-2026-11-09.md` in the public repository.

Citation strings such as `opencode-deepseek-v4.1-flash-free:KB-001` are ids of synthetic knowledge-base articles. They are not a live model call on the demo page.

Cases (also the backup if the live page is down)

If the live page is down, these are the same cases the suite already checks. The results below are measured, not guessed.

Case	Ticket	What to look at
Failure, locked	G-001, patient data in the ticket	Handoff. The model never sees the number or the name. Masking is Luhn-strict. The number is Skatteverket's published test number `19121212-1212` and becomes `[personnummer-maskat]`. The synthetic name becomes `[namn-maskat]`. Reasons: `patient_data_detected`, `model_abstained`, `low_confidence`, `no_grounded_draft`. Godkänn stays disabled.
Login draft	ticket-001, "Jag kan inte logga in till SITHS-systemet"	`awaiting_approval`. Citation `opencode-deepseek-v4.1-flash-free:KB-001`. A person still has to approve it.
Bad citation	ticket-008, save-button crash	Status is `awaiting_approval`, but the retrieved article is the print-queue runbook (`opencode-glm-5.3-flash-free:KB-002`). Faithful here means the draft is the retrieved article, not that the article is the right one. Reject it.
Abstain	G-008, compare two care episodes side by side	Not in the KB. No grounded draft, so handoff. Inventing a KB answer was rejected so this stays a miss.
Report, not an incident	G-009, avvikelsestatistik per enhet	Rules label `instruktion` and skip the model. No KB hit, so status is `handoff` with `no_grounded_draft`. The ticket flag `needs_human` stays false. The word avvikelse must not steal a statistics request. The route is first line, not `dev_team`.

A related case, G-003: a Luhn-invalid samordningsnummer in a patient context still hands off, and the masker still refuses to mask the number. The adjacent synthetic name is masked. That is why a second opinion exists.

How to run it

From the public repository:

```
git clone https://github.com/MarkusIsaksson1982/cambio-support-copilot.git
cd cambio-support-copilot
python3 -m venv .venv && .venv/bin/pip install -e '.[dev]'
.venv/bin/python -m pytest -q
.venv/bin/uvicorn copilot.api:app --host 127.0.0.1 --port 8000
```

Then open http://127.0.0.1:8000/

Hero check: run ticket G-001 from `data/gap_tickets_2026-11-05.jsonl` with `--adapter none`. Expect `status: handoff` and `patient_data_detected`. Patient data forces a human. The digits `19121212-1212` and the synthetic name must not appear in the envelope.

Recording

A silent screen recording of the demo page is in the repository at `docs/cambio-demo-recording.mp4` (G-001 handoff with masked personnummer and name, then the SITHS approval, then the rejected print-runbook citation, then the offline sidebar). This note and the offline pytest report are the other fallbacks.